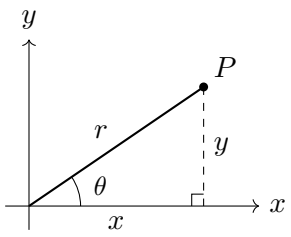
Review: Polar Coordinates

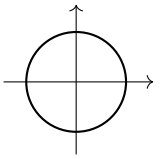
A point P in the plane can be located by its distance r from the origin and the angle θ measured counterclockwise from the positive x-axis. The right triangle in the figure gives the conversion between (x, y) and (r, θ). Throughout this section we take $r \geq 0$.



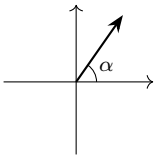
Converting coordinates.

$$x = r\cos\theta, \qquad y = r\sin\theta, \qquad r^2 = x^2+y^2, \qquad \tan\theta = \frac{y}{x}$$

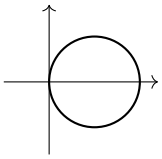
The curves below are the ones that make polar coordinates worth using. The last two are circles that pass through the origin; you can check them by multiplying both sides by r.



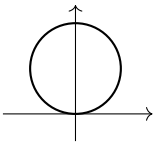
$r = a$
circle, center $(0, 0)$



$\theta = \alpha$
ray from the origin



$r = 2a\cos\theta$
circle, center $(a, 0)$



$r = 2a\sin\theta$
circle, center $(0, a)$

Example 1.

(a) Write the equation $x^2 + y^2 = 2x$ in polar coordinates and identify the curve.

(b) Write the equation $r = 6\sin\theta$ in Cartesian coordinates and identify the curve.

Why Polar?

Suppose we want the volume under the paraboloid $z = 4 - x^2 - y^2$ and above the xy-plane. The region underneath is the disk $x^2 + y^2 \le 4$. Using the methods of 5.2, it is a Type I region:

$$\iint_D (4 - x^2 - y^2)\, dA = \int_{-2}^{2} \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} (4 - x^2 - y^2)\, dy\, dx.$$

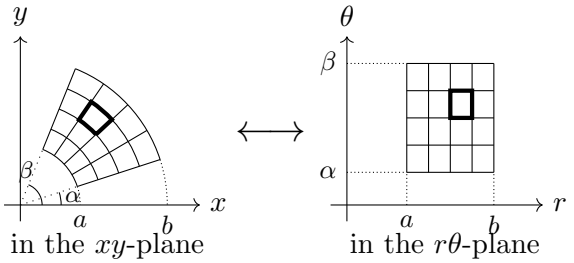
This integral is correct, but the square-root bounds make it unpleasant to evaluate. The disk is simple to describe in polar coordinates: $0 \le r \le 2$ and $0 \le \theta \le 2\pi$. Both variables run between *constants*, just as x and y do for a rectangle. That is the idea of this section.

Polar Rectangles

A **polar rectangle** is a region of the form

$$R = \{(r, \theta) : a \le r \le b, \quad \alpha \le \theta \le \beta\}.$$

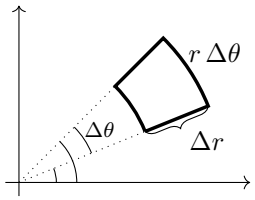
In the xy-plane it is a piece of a ring, but in the $r\theta$-plane it is an ordinary rectangle. As in 5.1, we chop R into small pieces: divide $[a, b]$ into subintervals of width Δr and $[\alpha, \beta]$ into subintervals of width $\Delta\theta$. One piece is drawn in bold at right.



What is the area of one piece? It is *not* $\Delta r\, \Delta\theta$. Pieces far from the origin are larger than pieces near it, even though every piece has the same Δr and $\Delta\theta$.

A sector of radius r and angle $\Delta\theta$ has area $\frac{1}{2} r^2 \Delta\theta$. The piece between radii r_{i-1} and r_i is the difference of two sectors:

$$\begin{aligned} \Delta A &= \tfrac{1}{2} r_i^2\, \Delta\theta - \tfrac{1}{2} r_{i-1}^2\, \Delta\theta \\ &= \tfrac{1}{2}(r_i + r_{i-1})(r_i - r_{i-1})\, \Delta\theta \\ &= r_i^*\, \Delta r\, \Delta\theta, \end{aligned}$$



where $r_i^* = \frac{1}{2}(r_i + r_{i-1})$ is the midpoint radius.
Geometrically, the piece is nearly a rectangle with one side Δr and the other side the arc length $r\, \Delta\theta$.
Adding up $f\, \Delta A$ over all the pieces and taking a limit gives the main result.

Double integral over a polar rectangle. If f is continuous on the polar rectangle $R = \{(r, \theta) : a \le r \le b,\ \alpha \le \theta \le \beta\}$, then

$$\iint_R f(x, y)\, dA = \int_{\alpha}^{\beta} \int_{a}^{b} f(r\cos\theta,\, r\sin\theta)\, r\, dr\, d\theta.$$

In short: substitute $x = r\cos\theta$, $y = r\sin\theta$, and replace dA with $r\, dr\, d\theta$. The extra r is not optional: it measures how much a small piece of the $r\theta$-plane is stretched when placed in the xy-plane. We will see it again in 5.7.

Example 2. Evaluate $\iint_R y\,dA$, where R is the region in the upper half-plane between the circles $x^2 + y^2 = 1$ and $x^2 + y^2 = 4$.

Practice 1. Evaluate each integral by converting to polar coordinates. Sketch the region first.

(a) $\iint_D \sqrt{x^2 + y^2}\,dA$, where D is the part of the disk $x^2 + y^2 \leq 9$ in the first quadrant.

(b) $\iint_D e^{x^2+y^2}\,dA$, where D is the disk $x^2 + y^2 \leq 1$.

Example 3. Sketch the region of integration, then convert to polar coordinates and evaluate:

$$\int_{-2}^{2}\int_{0}^{\sqrt{4-x^2}} (x^2+y^2)\,dy\,dx.$$

Practice 2.

(a) Use polar coordinates to find the volume of the solid under the paraboloid $z = 4-x^2-y^2$ and above the xy-plane. (This is the integral from *Why Polar?*.)

(b) Sketch the region of integration, then convert to polar coordinates and evaluate:

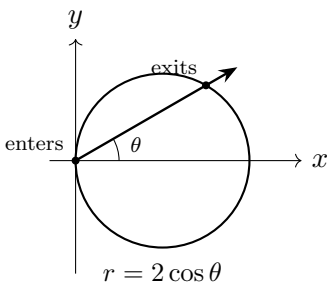
$$\int_{0}^{1}\int_{0}^{\sqrt{1-y^2}} \cos\left(x^2+y^2\right)\,dx\,dy.$$

General Polar Regions

In 5.2, a Type I region let the inner bounds on y be functions of x. Polar regions work the same way: the outer bounds on θ are constants, and the inner bounds on r may be functions of θ.

General polar region. If $D = \{(r, \theta) : \alpha \le \theta \le \beta,\ h_1(\theta) \le r \le h_2(\theta)\}$, then

$$\iint_D f(x, y)\, dA = \int_\alpha^\beta \int_{h_1(\theta)}^{h_2(\theta)} f(r\cos\theta,\, r\sin\theta)\, r\, dr\, d\theta.$$



To find the bounds, use the arrow method from 5.2, with the arrow now a *ray from the origin*.

- The ray enters the region at $r = h_1(\theta)$ and exits at $r = h_2(\theta)$.
- The θ-bounds are the first and last angles at which the ray hits the region.

Example 4. Let D be the region inside the circle $x^2 + y^2 = 2x$ (from Example 1). Evaluate

$$\iint_D \sqrt{x^2 + y^2}\, dA.$$

Answers. **Practice 1:** (a) $\dfrac{9\pi}{2}$ (b) $\pi(e-1)$ **Practice 2:** (a) 8π (b) $\dfrac{\pi \sin 1}{4}$

Example 4 (if not finished in class)**:** $\dfrac{32}{9}$